

CHUPUNGNYEONG, Korea, Republic of

WMO: 471350

Lat: 36.2203N

Lon: 127.9944E

Elev: 807

StdP: 14.27

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 13.3	(c) 16.5	(d) -4.4	(e) 4.5	(f) 21.3	(g) -0.9	(h) 5.4	(i) 23.7	(j) 25.1	(k) 26.3	(l) 22.2	(m) 24.6	(n) 9.2	(o) 270	(p) 0.411

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 13.6	(c) 89.7	(d) 75.1	(e) 87.3	(f) 74.1	(g) 84.9	(h) 72.8	(i) 77.2	(j) 85.7	(k) 76.2	(l) 84.1	(m) 75.1	(n) 82.5	(o) 5.6	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.9	134.8	81.6	73.8	129.8	80.4	72.8	125.4	79.5	41.4	85.9	40.3	84.4	39.3	82.7	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 19.0	(b) 16.5	(c) 14.4	(e) 8.3	(f) 92.7	(g) 3.7	(h) 2.0	(i) 5.6	(j) 94.2	(k) 3.4	(l) 95.4	(m) 1.3	(n) 96.5	(o) -1.4	(p) 98.0		
			DB													
			WB													
				8.3	92.7	3.7	2.0	5.6	94.2	3.4	95.4	1.3	96.5	-1.4	98.0	
				6.4	79.2	3.5	1.4	3.9	80.2	1.8	81.0	-0.1	81.9	-2.7	82.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.8	28.9	33.2	42.3	53.6	62.9	70.2	75.9	76.4	67.4	56.6	44.5	32.6		
	DBStd	17.47	6.36	6.60	6.62	6.51	4.73	4.14	4.67	4.58	4.99	5.74	6.89	6.70		
	HDD50	2152	654	471	250	35	0	0	0	0	0	13	190	539		
	HDD65	5073	1119	890	702	345	96	7	1	0	30	263	616	1004		
	CDD50	3551	0	1	13	143	401	605	803	819	522	219	25	0		
	CDD65	997	0	0	0	3	32	163	338	355	102	4	0	0		
	CDH74	7289	0	0	1	69	398	1087	2495	2696	523	20	0	0		
	CDH80	2086	0	0	0	5	57	234	806	902	82	0	0	0		
(14) Wind	WSAvg	5.7	8.1	7.5	7.4	7.0	5.8	4.4	3.9	3.8	3.8	4.4	5.7	7.1		
(15) Precipitation	PrecAvg	45.1	0.9	1.3	2.2	3.3	3.3	5.6	10.2	8.5	5.1	1.9	1.9	0.9		
	PrecMax	63.2	3.4	4.1	6.7	9.5	7.6	18.3	21.3	18.6	14.9	5.3	5.6	2.4		
	PrecMin	30.0	0.0	0.0	0.3	0.9	0.5	0.4	3.1	1.9	0.2	0.0	0.4	0.0		
	PrecStd	8.7	0.8	1.1	1.3	2.1	1.8	3.5	4.3	3.9	4.0	1.4	1.3	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.6	57.8	69.2	80.0	84.8	88.3	92.5	92.7	85.8	76.7	68.4	53.9		
		MCWB	42.0	48.6	52.0	59.4	61.8	68.4	76.7	76.0	72.5	61.4	56.0	45.3		
	2%	DB	44.9	52.5	63.5	75.7	81.3	85.1	89.7	90.1	82.4	72.9	63.3	49.6		
		MCWB	38.0	42.4	48.1	56.6	61.2	67.6	76.0	75.6	70.4	60.7	52.9	41.9		
	5%	DB	41.6	48.3	59.4	71.8	78.5	82.8	87.3	87.9	79.5	70.0	59.5	46.3		
		MCWB	35.1	39.3	45.8	54.4	59.8	66.9	75.1	74.9	68.6	59.2	51.0	39.7		
	10%	DB	38.5	44.4	55.3	67.7	75.5	80.3	84.8	85.3	77.0	67.2	55.9	43.2		
		MCWB	33.1	36.6	44.3	52.4	59.0	66.3	74.2	74.6	67.2	57.9	48.8	37.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.1	51.9	54.4	64.5	68.1	73.8	78.7	79.2	75.2	66.9	59.4	47.2		
		MCDB	47.3	56.2	61.6	72.3	74.1	81.3	88.2	87.9	81.8	71.4	64.8	52.3		
	2%	WB	38.9	43.9	51.0	60.1	65.0	71.9	77.3	77.5	73.2	63.9	55.1	43.4		
		MCDB	43.6	50.3	59.5	70.8	74.9	79.5	86.3	85.6	79.3	68.8	60.3	48.0		
	5%	WB	36.0	40.2	47.7	57.0	63.1	70.4	76.3	76.4	71.2	61.5	52.3	40.6		
		MCDB	40.2	46.7	57.4	67.9	73.0	77.8	84.6	84.3	76.1	67.3	58.0	45.6		
	10%	WB	33.6	37.5	44.9	54.4	61.2	68.9	75.2	75.5	69.2	59.2	49.7	38.1		
		MCDB	37.8	43.7	53.8	64.1	71.1	76.5	82.8	82.9	74.7	65.7	55.3	42.8		
(35) Mean Daily Temperature Range	MDBR		14.2	16.5	19.4	21.7	21.1	17.2	13.1	13.6	16.3	19.4	17.7	14.9		
	5% DB	MCDBR	18.6	23.0	27.6	29.6	27.8	22.7	17.5	18.4	19.7	22.9	23.5	20.5		
		MCWBR	12.7	14.8	15.2	13.9	11.4	8.7	6.6	6.6	9.0	11.4	13.9	13.9		
	5% WB	MCDBR	16.6	20.7	23.5	24.1	21.3	16.4	15.2	14.9	14.7	17.1	19.5	19.1		
(40) Clear-Sky Solar Irradiance	taub		0.400	0.444	0.523	0.555	0.564	0.623	0.579	0.536	0.466	0.443	0.423	0.396		
	taud		2.036	1.959	1.800	1.781	1.800	1.734	1.911	2.022	2.162	2.096	2.068	2.069		
	Ebn at Noon		239	243	235	237	237	222	232	239	250	242	229	231		
	Edh at Noon		41	50	64	69	69	73	61	54	44	43	40	37		
(44) All-Sky Solar Radiation	RadAvg		825	1056	1372	1574	1768	1605	1361	1384	1253	1139	828	736		
	RadStd		49	93	133	123	161	130	184	189	141	97	99	51		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	-2.34	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+135	N/A

Nomenclature: See separate page